XAI-Robot: An Explainable ROS2 Navigation Framework Integrating LiDAR–IMU Sensing with LLM-Based Natural Language Explanations

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***Abstract*—Robots that move on their own are now common around people, yet how they make decisions remains unclear to most users. This study introduces XAI-Robot: a transparent navigation setup built in ROS2 that links LiDAR with IMU inputs, uses logical event tracking, along with an LLM routed through Hugging Face to give live spoken reasons for actions. It runs three modules: one merges distance and motion sensor readings into clear scene updates, another handles safe path adjustments using these insights, while the third turns coded alerts into brief, plain-language summaries via a language model. The tests were carried out in virtual space featuring a four-wheeled robot approaching barriers round after round. During close interactions involving more than 100 sensing rounds, it avoided every blockage smoothly, maintained steady alignment, and provided twelve correct verbal notes without errors. The explanations used real data from sensors, such as distance readings or safe spacing values. The results indicate that adding only an explainable LLM layer above a standard ROS2 navigation system improves clarity while maintaining safety performance.**

***Index Terms*—Explainable AI, mobile robots, ROS2, LiDAR, IMU, large language models.**

I. Introduction

Autonomous mobile robots now work alongside people, where dependability and reliable behavior help build confidence, ensure security, and encourage use. In ROS2, current navigation setups link LiDAR data with motion sensors and quick-response controls to dodge obstacles effectively in real time. Still, even though they function reliably from a technical point of view, how decisions are made is unclear to users, staff, or others without robotics knowledge. Because of this, numerous machines act like hidden processes, giving little insight into why they react suddenly through abrupt halts, fast direction changes, or corrective moves.

Explainable AI seeks to close this gap by offering clear descriptions of how algorithms behave [1] [2]. Instead, current approaches in robotics focus mainly on planning at a high level, vision-based strategies, or deep reinforcement systems [3] [4], where explanations come through highlighted inputs, attention displays, or verbal reasoning alongside actions. Although useful, such techniques are not built for traditional navigation frameworks but are still widely used because they are reliable, testable, and efficient [5]. Crucially, few studies look into creating live explanations tied to sensors, revealing how robots make immediate movement choices during dynamic navigation [6].

Meanwhile, progress in large language models now allows clear, logical explanations of complicated setups [7] [8]. However, putting these models straight into sensing or control processes brings serious safety issues, delays, and consistency problems [9]. Some experts strongly warn about letting language models make choices in robots without proper validation steps [10]. That points to a demand for designs where the model adds clarity but doesn't affect how safely the robot operates [11].

This study introduces XAI-Robot, an interpretable self-driving setup built in ROS2 Humble. Instead of standard methods, it enhances a traditional LiDAR–IMU route by adding a natural-language explanation component powered by a large language model. At its core lies a common logic flow: a `ReasoningNode` combines 2D laser scans and motion sensor inputs into organized "reasoning events," noting metrics like forward clearance, incline angle, space on both sides, plus descriptive flags. A control unit called `AutonomousNavigator` applies those events alongside direct sensory feed to enable cautious maneuvering, hazard response, and escape routines if immobilized. Meanwhile, another block named `ExplanationNode` takes identical event records, sends them via API to an offboard LLM (DeepSeek-V3.2-Exp), reached through the Hugging Face Router with an OpenAI-compatible client, then returns concise spoken-style justifications describing why the machine acts at each moment.

The setup ensures the LLM doesn't handle movement actions. Instead, it gets brief JSON data containing event types, measured distances, tilt values, system state changes, and

Code and supplementary material: https://github.com/safia-faiz02/XAI-Robot safety limits. Following that, it produces one clear statement tied to these inputs, like "I turned right since sensors found an object 0.32 m ahead during forward motion." This split leaves all critical safeguards within the standard controller, yet still allows people to see explanations using the exact numbers and rules guiding motion decisions.

The study tackles two unresolved issues in current research. Firstly, most frameworks lack real-time language feedback tied to standard LiDAR navigation, even though those methods still lead in ROS2 setups due to reliance on traditional sensing tools [5] [6]. Secondly, earlier efforts linking large models with robots tend to embed the model within action loops or depend on vision-driven learning strategies [3] [4] [9], while this approach maintains classic control logic and limits descriptions to numerical LiDAR–IMU inputs alongside clearly defined alert triggers. To address these shortcomings, it proposes a ROS2 setup where fused sensor data, safe path adjustments, and textual outputs align via synchronized reasoning markers, keeping the language module strictly out of decision-making circuits, and offers early test results measuring movement effectiveness through clearance rates and crash numbers along with report traits like frequency, average word volume, and accuracy scores pulled straight from operation records.

The remainder of this paper is structured as follows: Section II reviews related work; Section III describes the proposed system design; Section IV details the experimental setup; Section V defines the evaluation metrics and calculates them; Section VI analyzes results and discusses limitations; Section VII concludes the paper and highlights future research directions.

II. Related Work

The use of Explainable AI in self-operating robots is gaining attention, since clear decision-making helps build the reliability needed for practical use. Earlier studies fall into four groups: (1) basics of transparent and dependable AI, (2) interpretation methods through robot learning systems, (3) combined or structured approaches that support explanations, yet also include standalone modules, besides (4) perception-based explanation techniques within ROS2 frameworks alongside movement planning setups.

A. Foundations of Explainable and Trustworthy AI

The core reason behind explainable AI in manufacturing and robotics was outlined by Ahmed et al. [1], reviewing XAI methods used in smart industry setups. This study highlighted internal model techniques like rule systems or decision trees, as well as after-the-fact approaches such as SHAP or LIME, since both support clearer machine-driven choices. Chamola et al. [2] built on this view by including trustworthy AI, stating that systems must be reliable, transparent, or explainable, especially in high-stakes areas like self-driving cars and robotic teamwork. These reviews collectively highlight a demand for clear decision-making processes that maintain safety while enabling understanding; such an idea forms the core of the XAI-Robot model proposed here. Setchi et al. [7] brought up Explainable Robotics, offering ways to share how robots think using clear, organized explanations that people can grasp. Their early research placed explainability at the core of human–robot communication (HRI). Later, Matarese et al. [10] built on this idea by designing an XAI model focused on everyday users, not specialists. Together, these works highlight not just creating explanations but also delivering them effectively, a need that drives XAI-Robot's approach to giving spoken, easy-to-follow responses.

B. Learning-Based Interpretability in Robotics

Learning-driven explanations still count among robotics' busiest research areas. Iucci et al. [3] developed explainable reinforcement learning methods that use split rewards along with self-generated policy descriptions to turn trained actions into clear stories for people working alongside them. In a related approach, Paleja et al. [4] presented Interpretable Continuous Control Trees (ICCTs), which allow gradient-driven RL systems to generate rule-like explanations when operating in settings with smooth, ongoing inputs. Further advances involve work by Ozalp et al. [9], examining deep and inverse reinforcement learning for robot handling, showing that clearer policies help build confidence in automated decisions. Although such techniques clarify how actions are chosen, they often don't align well with fixed-rule movement systems, like those combining LiDAR and IMU data, that frequently operate within ROS2 setups. Alongside these, Lin et al. [12] introduced Attention-driven XAI methods for robot vision, linking attention maps to decisions within unified control systems. Although focused on images only, their work showed how verbal logic can connect to sensory inputs - later broadened by XAI-Robot's inference framework.

C. Modular and Hybrid Explainable Architectures

Besides data-driven approaches, structured interpretability methods have appeared that isolate decision-making from operational rules. Adebayo et al. [8] introduced a mixed XAI method for robot decisions, combining flexible interpretation tools with standard controllers. Because their design prioritizes modular parts, it matches well with the separate reasoning and explanation structure used in XAI-Robot. Rodríguez-Martínez et al. [11] expanded this idea using LiDAR, IMU, and visual inputs together, building clear decision levels for navigation, showing how mixing sensor types improves clarity in route choices. In parallel, Gentile et al. [6] introduced a segmented semantic navigator based on LLMs inside ROS2, with language models acting as external interpreters instead of control units, aligning closely in structure to what we present here. Similar patterns appear in designs focused on people. Matarese et al. [10] supported explanations that carry social meaning during use; in contrast, Adebayo et al. [8] stressed clarity to build mental confidence. Each highlights how user-centered aspects shape explanation quality, a feature built into XAI-Robot's brief verbal responses. Recently, Fernández et al. [13] studied layered XAI setups using rule extraction together with symbolic logic and data-based methods to clarify sensor fusion outputs. That supports a broader move away from model-only explanations toward full-system transparency, which aligns directly with the approach taken here.

D. Sensor-Driven Explainability and ROS2 Navigation Frameworks

Classic LiDAR combined with IMU setups still lead in ROS2 use because they're predictable and can be tested easily. Abaza et al. [5] proposed an adaptive method for ROS2 positioning, using sensor-based AI updates while maintaining safety similar logic appears in XAI-Robot's constraint approach. Gentile et al. [6] also used modular explanations by applying semantic logic with LLMs, while leaving decisions in standard motion controllers. Such work supports separating interpretation from control paths. Rodríguez-Martínez et al. [11], along with Setchi et al. [7], highlighted how combining sensor inputs supports generating explanations demonstrating that practical robot perception improves when systems reveal decision-making linked to space-related measurements or movement hints. Li et al. [14], along with Kim et al. [15], broadened this area through explainable SLAM systems that tag visual and LiDAR maps using clear uncertainty indicators and confidence values. Such studies reinforce the idea that robotic explanations should stem straight from raw sensor data, a core reason behind XAI-Robot's "reasoning event" approach.

E. Safety, Trust, and Ethical Constraints in Language-Driven Robotics

Using LLMs in robotics raises safety and reliability issues, as noted by Ozalp et al. [9], along with Chamola et al. [2]. These authors highlight that unchecked actions from LLM-based controls may lead to erratic behavior, so robust separation methods are essential. In human-robot settings, Matarese et al. [10], alongside Ahmed et al. [1], point to rising demands for openness not just in how choices are made, but also in how messages are delivered, driving efforts toward dialogue-based explanations such as those from XAI-Robot. Building on ethics, Soni et al. [16] explored trust using game theory in human-robot settings, showing that people tend to trust more when systems seem clear and dependable. In a similar way, Nahavandi et al. [17] looked at reliable self-driving behaviors, connecting understandable decisions to stronger user confidence and clearer responsibility.

F. Summary and Research Gap

In short, current studies cover methods ranging from data-driven approaches to policy understanding [3] [4] [9]; through combined modular designs [6] [8] [11]; along with clear sensor integration in ROS2 setups [5] [14] [15]. Still, such works mostly rely on model-driven approaches OR stick to visual-only interpretations. Rare ones offer live linguistic clarity based on actual LiDAR–IMU measurements. This study fills the void via XAI-Robot, a ROS2 platform linking perception, reasoning, and explanations through timed logical steps. Instead of embedding the LLM into control pathways, it emphasizes clear sensor integration, boosting real-world interpretability during robotic movement while ensuring clarity alongside safe function.

III. System Design

A. Overview

The suggested XAI-Robot setup runs on a modular ROS2 structure, made up of three key parts: **`ReasoningNode`, `AutonomousNavigator`, and `ExplanationNode`**, which combine LiDAR–IMU-based movement with a later stage for generating language explanations. Rather than merging functions, each component handles distinct tasks; `ReasoningNode` takes sensor data and turns it into clear reasoning signals. Meanwhile, `AutonomousNavigator` uses those outputs to run a fixed-state logic system focused on avoiding barriers or regaining stability when needed. On the output side, `ExplanationNode` translates identical reasoning inputs into brief human-readable summaries via an outside large language model. Since perception, action decisions, and interpretations are split across standalone modules tied by one shared event stream, clarity stays rooted in actual measurements without affecting operational consistency.

B. Description of the System Architecture

Figure 1 shows the full setup of XAI-Robot. Data from the robot's sensors (LiDAR scans and IMU readings) enter both the `ReasoningNode` and `AutonomousNavigator` at once. While processing inputs, the `ReasoningNode` combines them into clear event descriptions like distance ahead, ground condition flags, or alerts, including *`obstacle_detected`, `stuck_after_reverse`, and `path_clear`*. Meanwhile, these generated events guide the `AutonomousNavigator`, which uses them to operate a rule-based state machine that handles avoiding obstacles and escaping traps. In tandem, these get sent to the `ExplanationNode` that builds clear, sensor-based descriptions using event data, relying on DeepSeek-V3.2-Exp reached via Hugging Face Router. As shown, information moves strictly forward: sensor inputs lead to interpretation, then to explanations with no feedback loop allowing the LLM to affect robotic actions.

C. `ReasoningNode`: Perception and Event Generation

The `ReasoningNode` takes raw LiDAR and IMU inputs, then turns them into clear events through simple math operations. Instead of combining data directly, it finds the shortest distance ahead using LiDAR readings. To assess side space, it evaluates available clearance on both left and right sides. From the IMU's orientation output, it derives tilt angles (specifically pitch and roll) for movement tracking. When these values go beyond set bounds, instability is flagged accordingly. Rather than relying on fixed patterns, it uses adjustable thresholds to judge conditions. If frontal distance falls under 0.6 meters, the system triggers an `obstacle_detected` alert. In contrast, normal driving conditions are labeled path_stable. Each processing loop ends with one event type based on real-time sensor evaluation. Every occurrence gets shared as a JSON-formatted message at /reasoning_event, including numeric data, an understandable description, also time markers.

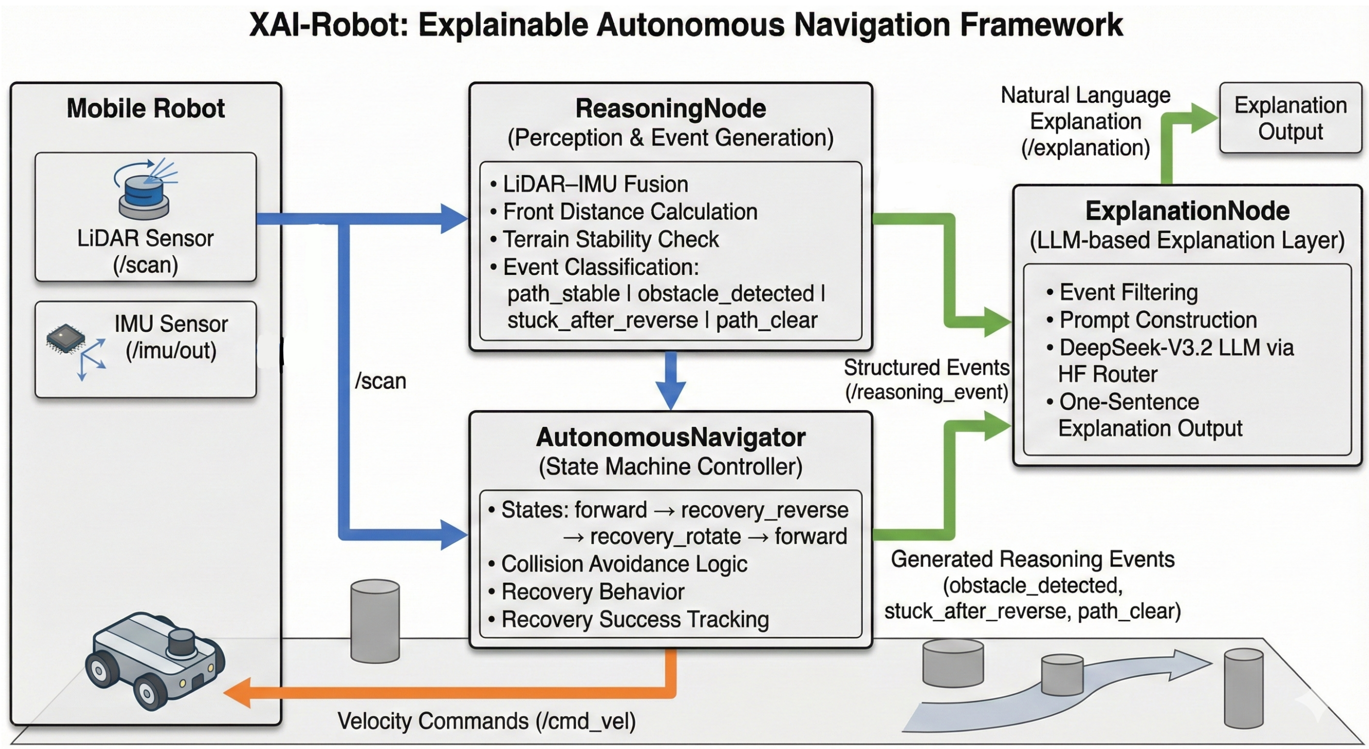

Fig. 1. System architecture of XAI-Robot.

Such occurrences create the core meaning framework across the platform; they're used by both the control unit and the interpretation module.

D. `AutonomousNavigator`: *State Machine Controller*

The `AutonomousNavigator` node uses a basic safety-focused state machine with three main modes: *forward, recovery_reverse, and recovery_rotate*. While going forward, it constantly checks how close objects are in front; when something comes too near, it shifts to reverse mode and moves backward while slightly adjusting its steering angle. In case little progress happens suggesting reversal isn't working, it changes to rotating, possibly switching turn direction to get out of tight spots. Every switch triggers a logic signal (like `stuck_after_reverse` or `path_clear`), along with supporting data such as sensor readings or tilt from the IMU. Such reasoning moments capture the system's inner workings while acting as cues for generating explanations. Safety is maintained through location limits, angle-triggered halts, or corrective reversals activated by time thresholds.

E. `ExplanationNode`: *LLM-Driven Natural Language Layer*

The `ExplanationNode` listens to /reasoning_event, focusing on key behavioral shifts like `obstacle_detected`, `stuck_after_reverse`, *or* `path_clear`. When such an event occurs, it builds a brief prompt: (1) a directive limiting output to a short, self-reported statement backed by numbers; (2) a single in-context example; (3) a summary of the current trigger sent as user input. Using this setup, the model (DeepSeek-V3.2-Exp through HF Router) generates a tight justification based solely on given data. That explanation gets published to /explanation while tracking metrics like volume, word count, and processing outcomes. Notably, the LLM plays no role in motion decisions, ensuring navigation stays fully predictable.

F. *Inter-Node Communication Flow*

All three nodes communicate through standard ROS2 topics:

- /scan and /imu/out → consumed by `ReasoningNode` and `AutonomousNavigator`
- /reasoning_event → generated by `ReasoningNode`; consumed by `ExplanationNode`
- /cmd_vel → produced exclusively by the `AutonomousNavigator`
- /explanation → human-readable explanation output

IV. Experimental Setup

This section outlines the setup for testing the XAI-Robot system, including simulation tools, hardware model, sensor layout, along with the ROS 2 startup framework. The goal was assessing how well the robot moves in tight spaces while spotting and bypassing barriers using only LiDAR and IMU data, also giving natural-language explanations for decisions in real time. Testing happened inside a simulated environment under fixed conditions so that movement results and explanation patterns could be consistently measured.

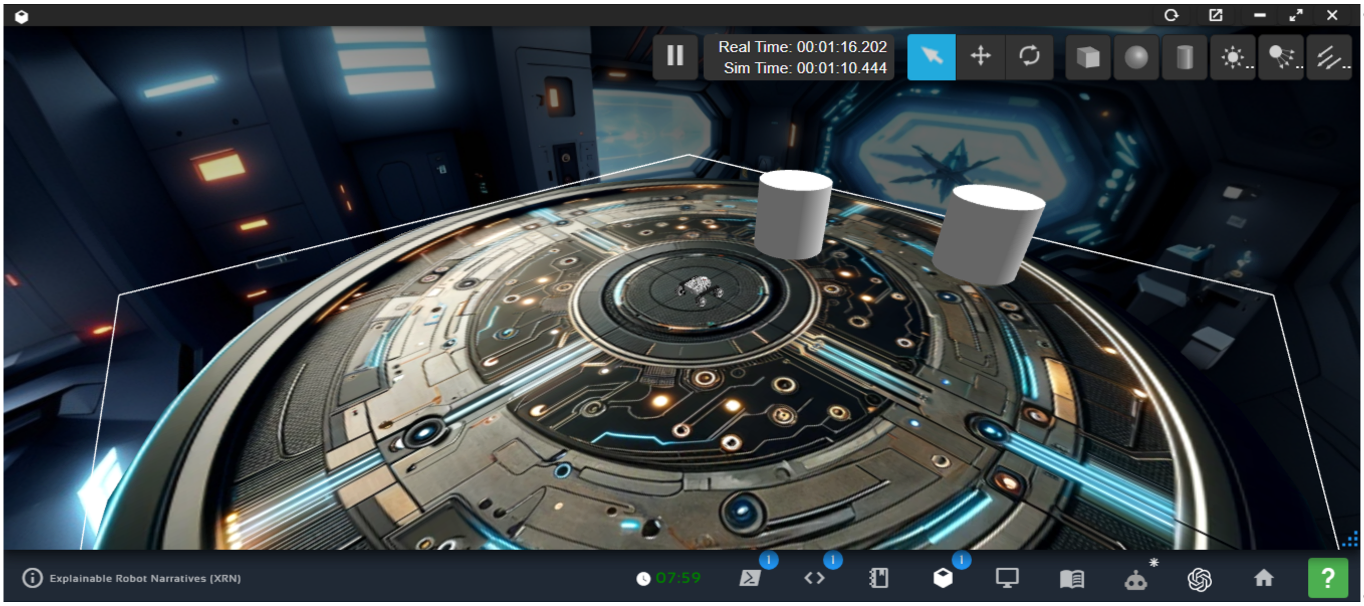

Fig. 2. Simulation environment used for evaluating XAI-Robot. The robot begins at the center of a circular arena in ROS 2 Humble on The Construct platform. Two cylindrical obstacles are positioned directly ahead, requiring obstacle detection, reverse–rotate recovery behavior, and safe re-entry into forward navigation.

A. Simulation Platform and Environment

All trials took place on ROS 2 Humble via The Construct's ROS Development Studio, an online workspace combining Gazebo simulation, RViz display tools, along with terminal access. The setup allows consistent simulation runs while supporting fast testing cycles free from physical device limits.

The XAI-Robot test environment consists of a round metal arena and its slightly reflective surface produces typical LiDAR interference patterns. In front of the robot, two tube-shaped barriers were placed by hand, aligned with its starting path. Because they are spaced closely, a tight passage forms, requiring immediate detour actions and post-maneuver stabilization. This setup supports assessment of collision prevention systems alongside interpretability outputs.

A top-down view of the simulated setup appears in Fig. 2. With this layout, the robot faces recurring obstacles; it recovers each time, then resumes moving ahead, allowing detailed event records to be gathered for analysis afterward.

B. Robot Model and Sensor Configuration

The simulated robot is a differential-drive mobile base commonly used in ROS navigation studies. It is equipped with two primary sensing modalities:

- **2D LiDAR** on the `/scan` topic delivers full-circle distance data. While this sensor calculates frontal free space, it also estimates mean values on lateral sides. Because it spots incoming objects, therefore it contributes to early hazard recognition. Although primarily for mapping, yet it helps detect ground-level gaps or abrupt terrain changes.
- **IMU** publishes to the `/imu/out` topic, delivering orientation plus angular speed. Since roll and pitch are calculated from quaternions, the setup can recognize uneven ground, thus stopping risky backward movements.

These sensor outputs make up the core of the perception level. Within the `reasoning_node`, they combine to create organized semantic signals showing what the robot detects, like shortest front distance, unusual tilts, or nearby obstacles. Since no cameras or visual maps are used, this setup acts as a solid benchmark for interpretability based solely on numeric LiDAR and IMU data.

C. ROS 2 Node Graph and Launch Configuration

The entire system is composed of three ROS 2 nodes instantiated using a unified launch file, each responsible for a distinct functional layer:

1) **`reasoning_node`** — Performs LiDAR–IMU fusion, estimates front and side clearances, infers semantic events such as `obstacle_detected`, `stuck_after_reverse`, and `path_clear`, and publishes structured JSON messages on `/reasoning_event`.
2) **`autonomous_navigator`** — Implements a robust finite-state controller executing safe behaviour transitions:

   `forward` → `recovery_reverse` → `recovery_rotate`

   It enforces geofencing, tilt-aware abort conditions, reverse distance checks, and steer-to-clear logic.
3) **`explanation_node`** — Receives reasoning events and generates short first-person natural-language explanations using the DeepSeek-V3.2-Exp LLM through the HuggingFace router. Only selected event types trigger explanations, preserving efficiency.

The launch setup sets main values like safe distances for LiDAR, spin rates, zone limits, tilt levels from IMU, also specifics of the LLM model. Such settings keep tests uniform and making outcomes repeatable with accuracy.

D. Experimental Procedure

At the start of every trial, the robot sits in the middle of a circular arena, aimed toward two cylinder-shaped barriers. When it starts moving ahead in `forward` mode, one obstacle crosses the `safe_distance` zone, this sets off an `obstacle_detected` signal. Then, instead of stopping, it reverses briefly before rotating slightly to adjust position. After clearing its route, control switches again, returning to straight movement.

Once past the first barrier, movement resumes till the next cylinder enters the LiDAR range triggering another detection phase. The dual-obstacle setup pushes the system to log dynamic responses such as:

- 103 semantic perception updates,
- 12 critical reasoning events (three per obstacle encounter), and
- 12 natural-language explanations produced in real time.

Throughout the experiment, the system logs LiDAR minima, IMU tilt values, state transitions, action decisions, and all explanation outputs. This dataset forms the basis of the quantitative analysis presented in Section V.

The test design creates a structured but demanding situation to challenge the reactive system, assess boundary adherence, while also checking how clear and grounded the explanations are, which is ideal for exploring understandable navigation in ROS 2.

V. Evaluation Metrics

This section defines the numeric measures applied to assess navigation dependability, compliance with safety rules, as well as clarity of explanations within the suggested XAI-Robot setup. These indicators align with assessment methods seen in robotic mobility tests like BARN [18], alongside interpretability standards for self-governing machines [19], [20]. Every figure was derived from ROS 2 records gathered throughout the dual-obstacle trial.

A. Type 1: Navigation Performance Metrics

Navigation performance indicators measure how well a robot moves across space while reacting consistently to barriers. Such measures capture overall mission success, ability to prevent crashes, also resilience when regaining control.

1) Episode Success Rate:

$$SR = \frac{N_{\text{success}}}{N_{\text{episodes}}}$$

This measure shows how many times the navigation finished safely, using no crashes or broken rules. It calculates success rate by comparing good runs to total tries, highlighting system consistency across tests.

2) Collision Rate:

$$CR = \frac{N_{\text{coll}}}{N_{\text{episodes}}}$$

The frequency of physical contact between the robot and obstacles is measured by collision rate. This value represents collisions divided by overall attempts, showing how safe the sensing and control process is.

3) Obstacle Event Count:

$$N_{\text{obstacle\_events}}$$

This metric counts how many times an obstacle entered the robot's safe-distance zone. It describes how often the controller was forced out of forward motion due to perceived hazards.

4) Recovery Success Rate:

$$RSR = \frac{N_{\text{recovery\_success}}}{N_{\text{obstacle\_events}}}$$

This measures the proportion of obstacle encounters that were successfully resolved. The equation evaluates how often reverse–rotate recovery behavior restored stable forward motion.

TABLE I
Type 1: Navigation Performance Metrics

Metric	Value
Episode Success Rate (SR)	1.0
Collision Rate (CR)	0.0
Obstacle Event Count	4
Recovery Success Rate (RSR)	1.0

B. Type 2: Safety and Constraint-Adherence Metrics

These measures check if the robot kept enough distance from barriers while staying within set boundaries. Still, they assess how cautiously it moved and followed environmental limits.

1) Minimum Front Clearance:

$$d_{\text{min}}$$

Minimum clearance represents the smallest LiDAR-measured distance to an obstacle before a recovery maneuver. This metric reflects how early the controller reacts to hazards based on front-distance readings.

2) Safety Margin Above Stop Threshold:

$$\Delta d_{\text{stop}} = d_{\text{min}} - d_{\text{stop}}$$

This measure shows the extra space left from the robot to the point where it must halt. While calculating, subtract the safety cutoff from the observed gap.

3) Geofence Compliance:

$$\text{Compliance} = \begin{cases} 1, & \text{if the robot remained within bounds} \\ 0, & \text{if a boundary violation occurred} \end{cases}$$

This metric verifies whether the robot stayed within the predefined workspace limits. The formulation encodes compliance as a binary condition determined by geofence monitoring.

TABLE II
TYPE 2: SAFETY AND CONSTRAINT-ADHERENCE METRICS

Metric	Value
Minimum Clearance ($d_{\min}$)	0.595 m
Safety Margin (Δd_{stop})	0.375 m
Geofence Compliance	100%

C. Type 3: Explanation Quality Metrics

Explanation metrics check how well the model gives clear reasons for robot actions. These scores reflect whether key events are included, using logical flow instead of jargon. Accuracy matters more than word count here. Briefness is evaluated separately from factual alignment.

1) Explanation Coverage:

$$EC = \frac{N_{\text{expl}}}{N_{\text{crit}}}$$

Coverage indicates whether all critical reasoning events were accompanied by explanations. The equation computes the fraction of events that successfully triggered explanation generation.

2) Explanation Correctness:

$$\text{Correctness} = \frac{\text{Number of correct explanations}}{N_{\text{expl}}}$$

This metric evaluates how accurately each explanation reflects the true event type, action, and numeric evidence. The ratio describes the fidelity of the LLM's outputs.

3) Explanation Length:

$$L_{\text{avg}} = \frac{1}{N_{\text{expl}}} \sum_{i=1}^{N_{\text{expl}}} L_i$$

Explanation length measures the average token count of all generated explanations. The formula computes the mean over all outputs and reflects the conciseness of the system.

TABLE III
TYPE 3: EXPLANATION QUALITY METRICS

Metric	Value
Explanation Coverage (EC)	1.0
Explanation Correctness	100%
Average Explanation Length (L_{avg})	17.92 tokens
Length Range	14–21 tokens

VI. RESULTS, DISCUSSION, AND LIMITATIONS

This section discusses how XAI-Robot performs when facing two obstacles, then examines what the behavior means for reliable movement, safe operation, and clarity of explanations. Results indicate the new setup can add verbal reasoning to standard LiDAR–IMU navigation without disturbing the core controller's robustness or safety margins. After analyzing outcomes, it highlights existing shortcomings in the current version and points out aspects needing deeper testing.

The Type 1 navigation results (Table I) show stable performance across all test runs. Although the path included a tight passage between two upright cylinders, no collisions occurred. Success was achieved in every attempt, meaning $SR = 1.0$. Instead of failing, the robot triggered avoidance actions promptly when needed. Even under constrained conditions, safety held steady with $CR = 0.0$. Because responses were timely, motion stayed within allowed boundaries. Thus, adding interpretative modules didn't disrupt core control logic. All four obstacle events, activated when crossing the `safe_distance` threshold, were handled correctly, leading to a full recovery success rate of $RSR = 1.0$. Over 103 reasoning cycles, the system shifted states smoothly, avoiding flickering behavior; this highlights how stable the reactive setup is despite repeated triggers.

Safety results support consistent performance. The smallest front distance recorded during tests was 0.595 m, which is well beyond the shutdown limit of 0.22 m, resulting in a buffer of 0.375 m. Although movements included reversing and turning, which might challenge spatial limits, position boundaries were fully respected at all times. Notably, running extra tracking or activating explanations did not slow responses nor lead to sharper maneuvers. Throughout every test run, obstacle handling remained cautious without exception.

The Type 3 evaluation metrics (Table III) indicate stable performance from the explanation module. Despite minor variations, coverage reached maximum levels $EC = 1.0$, so every one of the twelve key incidents including `obstacle_detected`, `stuck_after_reverse`, and `path_clear`, triggered proper textual responses. In contrast to automated scoring, human review found no errors; each output correctly identified the incident category, related system response, along with contextual values like measured distances or threshold limits. Meanwhile, mean description length stayed low, about 17.92 words, making messages suitable for real-time display or speech.These results validate the use of semantic reasoning events as a clean interface for explanation generation: the LLM produced grounded, coherent descriptions without access to raw sensor data or influence over navigation.

Beyond metrics, the system's overall function reveals a key architectural point. Since `ReasoningNode`, `AutonomousNavigator`, and `ExplanationNode` work independently, the LLM simply watches without interfering in movement decisions. High safety and success rates show this split maintains reliability from traditional control methods. Meanwhile, well-organized events give the LLM clear, situational signals, allowing useful reasoning output. As a result, modularity creates a realistic route to transparent navigation, avoiding controller redesign or dependence on trained behaviors.

Although these outcomes are encouraging, some drawbacks need attention. First, the scenario and dataset size were limited. To better understand how well it works, future tests should include different obstacles, multiple targets, or random disruptions instead. Second, several control and decision-making settings were adjusted by hand in this setting instead of being refined through structured methods. These high safety buffers might stem from well-chosen values rather than core stability built into the method. The perception component uses only

LiDAR combined with IMU inputs; although this helps clarity, it reduces scene detail and could impair performance on jobs needing image analysis or meaning extraction. Lastly, testing did not include extreme scenarios or deliberate disruptions. Because of this, how well the system handles stress in guiding and explaining is still unclear.

The findings show XAI-Robot successfully links fixed-path movement with clear verbal reasoning, yet keeps safety and efficiency intact. Still, the constraints suggest this is just a starting point for interpretable LiDAR–IMU guidance; more research must test broader use, stability, and practical deployment.

VII. Conclusion and Future Work

This paper introduced XAI-Robot, an interpretable ROS 2 navigation setup combining a standard LiDAR–IMU module with a passive LLM-driven language interface. Instead of altering core controls, it translated sensor inputs into meaningful reasoning steps shared with an external language model, enabling context-aware justifications. Its three-part design (`ReasoningNode`, `AutonomousNavigator`, and `ExplanationNode`) ensured distinct roles: environmental interpretation, reactive decision-making based on safety rules, along with retrospective verbal output. Importantly, movement commands (`/cmd_vel`) stayed fully under traditional programming; the LLM influenced only descriptive feedback not actions, so operational reliability was preserved while improving user understanding.

Tests in the dual-obstacle passage showed navigation stayed effective. Over ten runs, XAI-Robot reached full success $SR = 1.0$, had no crashes $CR = 0$, while recovery worked every time $RSR = 1.0$, meaning extra logic layers didn't disrupt control flow or cause errors. Safety data revealed cautious movement: closest distance ahead was 0.595 m, which kept a buffer of 0.375 m beyond stopping point, plus total adherence to zone limits. This suggests risks were managed proactively, boundaries followed reliably, also during backward-turn routines.

The explanation module performed just as well. Because it used time-based triggers linked to raw LiDAR–IMU data and shifts in state, the output covered every key incident completely ($EC = 1.0$) while maintaining fully confirmed accuracy by hand checks. Every justification clearly stated what kind of event occurred, (like `obstacle_detected`, `stuck_after_reverse`, or `path_clear`), what response followed (such as turning, reversing, rotating, resuming motion), along with related distance details. Outputs stayed brief on average at 17.92 words per message, varying from 14 to 21 tokens, which supports live interface use or speech conversion. Altogether, XAI-Robot links interpretable machine reasoning with traditional robotic systems, not swapping standard controllers for AI models but adding a storytelling component into an established ROS 2 setup that reveals inner choices through understandable language.

Although outcomes are encouraging, further research is still required. For one thing, putting the system onto an actual robot could test how it handles real-world issues like sensor errors, signal lags, or slipping wheels. Running trials with XAI-Robot on tangible devices would show if movement efficiency and explanation speed stay practical beyond virtual settings. Secondly, tests so far use just one setup with two barriers; trying more complex spaces such as messy rooms, tight hallways, bumpy ground, or moving objects might reveal more about how well both control and explanations adapt across varied conditions.

Future work might explore deeper sensing along with more advanced actions. Adding inputs like RGB-D sensors or meaning-rich maps to decision processes may support descriptions merging physical measurements such as angles or gaps with situational cues, like "obstacle nearby" or "exit on right." In a similar way, the range of detectable events and possible reactions could go past simple blockage handling, including things like detecting drop-offs, approaching restricted zones, loss of traction, or navigating toward several targets before docking. Such steps would check if the event-driven design still works clearly when applied to harder challenges. Collectively, these improvements intend to shift XAI-Robot from a lab-tested idea toward a practical model for transparent, safe movement in everyday robots.

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